

CSA0614

## DESIGN AND ANALYSIS OF ALGORITHMS

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### Experiment 16: Selection Sort

#### Code:

```
def top_k_scores(arr, k):  
    arr = arr.copy()  
    n = len(arr)  
    for i in range(min(k, n)):  
        max_index = i  
        for j in range(i + 1, n):  
            if arr[j] > arr[max_index]:  
                max_index = j  
        arr[i], arr[max_index] = arr[max_index], arr[i]  
    return arr[:k]  
  
scores = [72, 88, 65, 90, 77, 95, 60, 83, 91, 68]  
print("Top 5 Scores:", top_k_scores(scores, 5))
```

#### Output:

```
=== RESTART: C:/Users/hk838/Download  
Top 5 Scores: [95, 91, 90, 88, 83]
```